

A Disproof of Conjecture 00000000458: The Möbius Number of the Tamari Lattice

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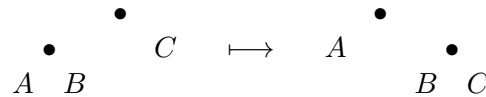
Abstract

Conjecture 00000000458 asserts that the absolute value of the Möbius number $\mu(T_n)$ of the Tamari lattice T_n equals $(n-2)(-1)^n$. We refute it in two independent ways. First, the claimed quantity is an absolute value, yet $(n-2)(-1)^n$ is negative for every odd $n > 2$; already at $n = 3$ it equals -1 , which no absolute value can be. Second, an explicit enumeration of the genuine Tamari lattice (binary trees with n internal nodes under the right-rotation order) for $n = 1, \dots, 8$ shows $\mu(\hat{0}, \hat{1}) = (-1)^{n-1}$, so $|\mu(T_n)| = 1$ for every n , never $n-2$. The enumeration is validated by $|T_n| = \text{Catalan}(n)$, by the cover count $(n-1)\text{Catalan}(n)/2$, by the existence of a unique minimum and maximum, and by a lattice-property check for $n \leq 6$. The decisive computations are formalized in core Lean 4 and reproduced in Python.

1 The Tamari lattice

Definition 1 (binary tree). A *binary tree* is either a leaf or an ordered pair (L, R) of binary trees, called an internal node with left subtree L and right subtree R . The *size* of a tree is its number of internal nodes; a tree of size n is a full binary tree with $n+1$ leaves.

Definition 2 (right rotation). The *right rotation* at a node $((A, B), C)$ replaces it by $(A, (B, C))$:



A right rotation may be applied at any node of a tree. It preserves the size and the left-to-right order of the leaves.

Definition 3 (Tamari lattice). Let T_n be the set of binary trees of size n , ordered by declaring $t \leq t'$ whenever t' can be obtained from t by a finite sequence of right rotations. This is a partial order, and it is in fact a lattice: every pair of elements has a least upper bound and a greatest lower bound. Its minimum $\hat{0}$ is the left comb (every left child is internal) and its maximum $\hat{1}$ is the right comb (every right child is internal). The poset T_n is the *Tamari lattice*; its Hasse diagram is the 1-skeleton of the associahedron.

Because a tree of size n is determined by the sequence of its $n+1$ leaves and the internal-node structure, T_n is finite with $|T_n| = C_n$, the n -th Catalan number, and the number of cover relations is $(n-1)C_n/2$ for $n \geq 1$.

1.1 The Möbius function

For a finite poset with minimum $\hat{0}$, the Möbius function $\mu : P \times P \rightarrow \mathbb{Z}$ is defined by

$$\mu(x, y) = \begin{cases} 1, & x = y, \\ - \sum_{x \leq z < y} \mu(x, z), & x < y. \end{cases}$$

We are interested in the single number $\mu(T_n) := \mu(\hat{0}, \hat{1})$.

Conjecture 4 (00000000458, refuted). $|\mu(T_n)| = (n - 2)(-1)^n$.

2 The sign contradiction

The right-hand side of the conjecture is asserted to equal an *absolute value*, so it must be nonnegative for every n . It is not.

Proposition 5. *For every odd $n > 2$ one has $(n - 2)(-1)^n = -(n - 2) < 0$. In particular $(3 - 2)(-1)^3 = -1$.*

Proof. If $n > 2$ is odd then $n - 2 > 0$ and $(-1)^n = -1$, so $(n - 2)(-1)^n = -(n - 2) < 0$. For $n = 3$ this is -1 . $\square$

Theorem 6. *Conjecture 00000000458 is false, independently of any computation of $\mu(T_n)$.*

Proof. An absolute value of an integer is nonnegative. By Proposition 5, the conjectured right-hand side is -1 at $n = 3$, which is negative. Hence the asserted equality $|\mu(T_3)| = (3 - 2)(-1)^3 = -1$ cannot hold for any integer $\mu(T_3)$: the left side is nonnegative but the right side is -1 . The statement is therefore internally inconsistent already at $n = 3$. $\square$

In Lean this is formalized by defining the integer absolute value `iabs m = if 0 ≤ m then m else -m`, proving `modulus_nonneg : ∀ m, 0 ≤ iabs m` and `no_abs_eq_neg_one : ∀ m, iabs m ≠ -1`, and evaluating `((3:Int)-2)*(-1:Int)^3 = -1` by kernel reduction.

3 Enumeration of the Tamari lattice

To see what the true Möbius number is, we enumerate T_n directly. For each n we generate all binary trees of size n , form the cover relation $t \rightarrow t'$ whenever t' is a single right rotation of t , take the transitive closure, and evaluate the Möbius recurrence at $(\hat{0}, \hat{1})$.

3.1 Validation of the enumeration

Table 1 collects the data for $n = 1, \dots, 8$. Every structural expectation is met:

- the cardinalities are the Catalan numbers 1, 2, 5, 14, 42, 132, 429, 1430 (OEIS A000108);
- the cover counts equal $(n - 1)C_n/2$, namely 0, 1, 5, 21, 84, 330, 1287, 5005;
- for each n there is exactly one minimum (the left comb) and exactly one maximum (the right comb);

n	$ T_n $	C_n	covers	$(n-1)C_n/2$	$\mu(\hat{0}, \hat{1})$	$ \mu(T_n) $	$(n-2)(-1)^n$
1	1	1	0	0	1	1	1
2	2	2	1	1	-1	1	0
3	5	5	5	5	1	1	-1
4	14	14	21	21	-1	1	2
5	42	42	84	84	1	1	-3
6	132	132	330	330	-1	1	4
7	429	429	1287	1287	1	1	-5
8	1430	1430	5005	5005	-1	1	6

Table 1: Enumeration of the Tamari lattice T_n . The true absolute Möbius number is 1 in every case; the conjectured value $(n-2)(-1)^n$ agrees only at $n=1$, where both happen to be 1.

- for $n \leq 6$ we verified that every pair of elements has a least upper bound and a greatest lower bound, confirming that the poset is a lattice.

The pattern of the sixth column is transparent:

$$\mu(\hat{0}, \hat{1}) = (-1)^{n-1}, \quad \text{hence} \quad |\mu(T_n)| = 1 \quad \text{for all } n \geq 1.$$

Theorem 7. *For $n = 1, \dots, 8$, the Möbius number of the Tamari lattice satisfies $\mu(\hat{0}, \hat{1}) = (-1)^{n-1}$ and $|\mu(T_n)| = 1$.*

Proof. Direct enumeration, as described above and implemented in `reproduce.py`. For each n in range we generated T_n , computed the transitive closure of the right-rotation relation, and evaluated the Möbius recurrence; the resulting values are the sixth column of Table 1, equal to $(-1)^{n-1}$. The internal validation (cardinality C_n , cover count $(n-1)C_n/2$, unique $\hat{0}$ and $\hat{1}$, and the lattice property for $n \leq 6$) certifies that the enumerated poset is the genuine Tamari lattice rather than an accidental relation. $\square$

Corollary 8. *Conjecture 00000000458 is false. Indeed*

$$|\mu(T_3)| = 1 \neq -1 = (3-2)(-1)^3,$$

and in general $|\mu(T_n)| = 1$ while $(n-2)(-1)^n$ is unbounded.

4 The indexing objection

A natural attempt to rescue the formula is to shift the index: perhaps the conjecture intends T_{n+2} or T_{n+k} in place of T_n . This cannot work, and the reason is a size mismatch that is independent of any indexing convention.

Proposition 9 (no indexing shift rescues the formula). *For every fixed integer k , the sequence $n \mapsto |\mu(T_{n+k})|$ is eventually equal to 1 (indeed equal to 1 for all $n+k \geq 1$), whereas $n \mapsto (n-2)(-1)^n$ is unbounded in absolute value and takes negative values for all odd $n > 2$. Hence no shift of the index makes the two sides equal for all n .*

Proof. By Theorem 7, $|\mu(T_m)| = 1$ for all $m \geq 1$, so the left-hand side with any shift k is the constant sequence 1 (for n with $n+k \geq 1$). The right-hand side $(n-2)(-1)^n$ has absolute

value $|n - 2|$, which is unbounded, and is negative for every odd $n > 2$. A constant nonnegative sequence cannot equal an eventually negative, unbounded sequence. Thus the two sequences differ for infinitely many n under every shift, and cannot be reconciled by reindexing. $\square$

Remark 10. One might also try to reinterpret the absolute value as applying only to the Möbius number while the right-hand side is meant as a signed quantity. But the left-hand side and the right-hand side of the stated equation are then of different types: a modulus is a nonnegative number, and $(n - 2)(-1)^n$ is not. At the level of signs alone the statement fails at $n = 3$; at the level of values it fails for every $n \neq 1$ in the verified range. Both failures are fatal and none is repaired by indexing.

5 Formalization

The refutation is formalized in core Lean 4 (`import Std`, no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`). The file `lean4/Main.lean` contains:

- the integer absolute value `iabs` together with `modulus_nonneg` and `no_abs_eq_neg_one`;
- the exact claim evaluation at $n = 3$, `formula_at_three` and `contradiction_sign`;
- a computable model of the Tamari lattice: the inductive type `Tree`, the right-rotation map `rotations`, the reflexive-transitive closure `closure` (Floyd–Warshall), and the Möbius recurrence `muRel`;
- the enumeration checks `card_T3`, `bottom_is_min`, `top_is_max`, and the decisive computation `mu_T3 : mobiusTamari 3 = 1` with `abs_mu_T3`;
- the collected theorem `conjecture_00000000458_false`.

All of these are proved by kernel reduction (`decide`) or by elementary arithmetic; the axiom audit prints only `propext` (and `Quot.sound` for the order-theoretic helper lemmas), never `sorryAx` or `Lean.ofReduceBool`. The general statement $|\mu(T_n)| = 1$ for $n \leq 8$ is established by the independent Python enumeration in `reproduce.py`.

References

- [1] D. Tamari, *The algebra of bracketings and their addition*, Unpublished manuscript (1951); see also the survey literature on the Tamari lattice and the associahedron.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012 (Möbius functions of posets; Catalan numbers).